

DELCODE Clinical Progression Data – Documentation

Note: All diagnoses provided in the April 2022 release of clinical progression data (“_untilApril21”) are based on the DELCODE release from April 2021.

1. Incident Mild Cognitive Impairment (MCI)

Clinical progression to incident MCI was assessed in all diagnostic groups, who were cognitively normal at baseline (i.e. controls, AD relatives, SCD).

Diagnostic Procedure

Each participant’s MCI status up until April 12th 2021 was determined in a two-step review process, which was adapted from the consensus conference process implemented in the Wisconsin Registry for Alzheimer’s Prevention^{1,2}. First, potential MCI converters were identified using a liberal “flagging algorithm”. More specifically, any participant, who showed a) cognitive test deficits, b) an abnormal informant report [CDR, FAQ], and/or c) a test performance that fell below specified cut-offs [in MMSE, CERAD delayed recall or FCSRT] at any follow-up visit, was flagged as a potential converter. Participants, who did not meet any of the flagging criteria, were classified as non-converters / cognitively normal after an additional review of their cognitive profiles by a neuropsychologist.

In the second review step, a consensus committee consisting of neuropsychologists examined those flagged as potential converters individually. Based on the cognitive and clinical data from all available study visits of the respective participant the committee examined, whether clinical progression to MCI had taken place or the participant remained cognitively normal at the most recent follow-up visit. Additionally, the subtype of impairment was specified for those who showed progression to MCI (i.e. single domain amnesic, multi domain amnesic, non-amnesic).

Consistent with the group definitions at study inclusion, an age, sex and education-adjusted performance of at least -1.5 SD was set as the threshold for an impaired test performance. A diagnosis of incident MCI necessitated a) a sufficient deficit in at least one cognitive domain, which was consistent across both cognitive tests and study visits, and b) evidence for cognitive decline either in longitudinal cognitive test performance or in self-/informant reports of cognition. Participants, who were flagged as potential converters in the first review step, but did not fulfill these criteria, were classified as non-converters / cognitively normal (e.g. only isolated test deficits, transient decline with subsequent improvement).

Exported Variables

Name	Description
MCI_untilApril21	Classification of clinical progression - 0: cognitively stable - 1: progression to MCI
MCIDomains_untilApril21	Classification of MCI subtype - 0: cognitively stable - 1: single domain amnesic - 2: multi domain amnesic - 3: non-amnesic
MCIDate_untilApril21	Date of first MCI diagnosis or censoring (YYYY-MM-DD) [excluding telephone interviews]
MCITime_untilApril21	Time difference baseline – event/censoring date (in years) [excluding telephone interviews]

2. Incident Dementia

Clinical progression to incident dementia was assessed in all diagnostic groups, who did not have dementia at baseline (i.e. controls, AD relatives, SCD, MCI).

Diagnostic Procedure

A diagnostic committee consisting of Prof. Dr. Jessen and members of the Neuropsychology research group reviewed all DELCODE cases with a documented dementia conversion, as well as cases without a dementia diagnosis, but a CDR global score of ≥ 1 and/or a MMSE total score of ≤ 23 . The aim was to identify cases, where a dementia conversion was potentially “missed”, as well as cases with a documented conversion, but potentially insufficient cognitive and/or functional deficits for a diagnosis of dementia. To that end, the committee reviewed all available data regarding the dementia diagnoses and clinical assessments of dementia progression provided by the study physicians, as well as CDR, MMSE, FAQ, and neuropsychological data.

If questions regarding the appropriate diagnostic classification of a participant or the type of dementia were identified, a further review of the respective case was requested from the physicians of its corresponding study center. The final diagnostic classifications and documented dementia types are based on the available clinical and diagnostic data as well as the query responses from the study physicians. If a study center review was requested, the diagnostic classification was based on the answers given by the study physicians.

Exported Variables

Name	Description
Dem_untilApril21	Classification of clinical progression - 0: no dementia - 1: progression to dementia
DemType_untilApril21	Clinically diagnosed dementia type - 0: no dementia - 4: Alzheimer’s disease dementia (probable or possible) - 7: mixed dementia - 8: vascular dementia - 19: Lewy body dementia - 99: frontotemporal dementia - 404: dementia type unknown
DemDate_untilApril21	Date of first dementia diagnosis or censoring (YYYY-MM-DD) [including telephone interviews]
DemTime_untilApril21	Time difference baseline – event/censoring date (in years) [including telephone interviews]

References

- Koscik, R. L., Berman, S. E., Clark, L. R., Mueller, K. D., Okonkwo, O. C., Gleason, C. E., . . . Johnson, S. C. (2016). Intraindividual Cognitive Variability in Middle Age Predicts Cognitive Impairment 8-10 Years Later: Results from the Wisconsin Registry for Alzheimer's Prevention. *Journal of the International Neuropsychological Society*, 22(10), 1016–1025.
- Koscik, R. L., Hermann, B. P., Allison, S., Clark, L. R., Jonaitis, E. M., Mueller, K. D., . . . Johnson, S. C. (2021). Validity Evidence for the Research Category, "Cognitively Unimpaired - Declining," as a Risk Marker for Mild Cognitive Impairment and Alzheimer's Disease. *Frontiers in Aging Neuroscience*, 13, 688478.